



Manufacturing Operations & Supplier Performance Report

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Target Audience

This dashboard suite is designed for manufacturing operations leaders, supply chain and procurement managers, and BI/data leadership who want clear, actionable visibility into production efficiency, product quality, and supplier performance.

Description of the Data

This platform is built on the AdventureWorks2025 sample database (Microsoft), transformed through a dbt pipeline into three fact tables and four supporting dimensions. Each fact table captures a different stage of manufacturing operations:

1. **Work Orders (June 2022 – June 2025)**
 - Source: AdventureWorks2025 Sample Database, modeled via dbt (fct_work_orders)
 - Units: Days (cycle time), Percent (scrap rate, on-time rate), Units (quantity produced)
 - Description: Work order–level production records covering order quantity, scrap quantity, cycle time, scrap reason, and on-time delivery status across product categories.
2. **Purchase Orders (June 2022 – June 2025)**
 - Source: AdventureWorks2025 Sample Database, modeled via dbt (fct_purchase_orders)
 - Units: Days (lead time), Percent (fulfillment rate), USD (spend)
 - Description: Purchase order–level procurement records covering vendor lead time, order completion status, and total spend by vendor.
3. **Product Vendor Pricing**
 - Source: AdventureWorks2025 Sample Database, modeled via dbt (fct_product_vendor_pricing)
 - Units: USD (standard price), Percent (price variance)
 - Description: Vendor- and product-level pricing records used to evaluate price variance across vendors supplying the same parts.
4. **Supporting Dimensions**
 - dim_product, dim_vendor, dim_scrap_reason, dim_date built via dbt to support consistent filtering, categorization, and time intelligence across all three fact tables.

Technical Approach

Unlike a purely visualization-based project, this platform was built as a full BI engineering pipeline:

- **Data Source:** AdventureWorks2025 sample database, hosted in SQL Server running in a Docker container.
- **Transformation Layer:** dbt (data build tool), implementing layered architecture staging models, dimension models, and fact models with automated data quality tests at each layer.
- **Visualization Layer:** Power BI Desktop, connected directly to the SQL Server instance, with a custom theme and consistent design system applied across both dashboards.

Purpose of the Dashboards

The purpose of this dashboard suite is to provide a clear, end-to-end view of manufacturing operations from the shop floor through the supply chain.

It allows the viewer to:

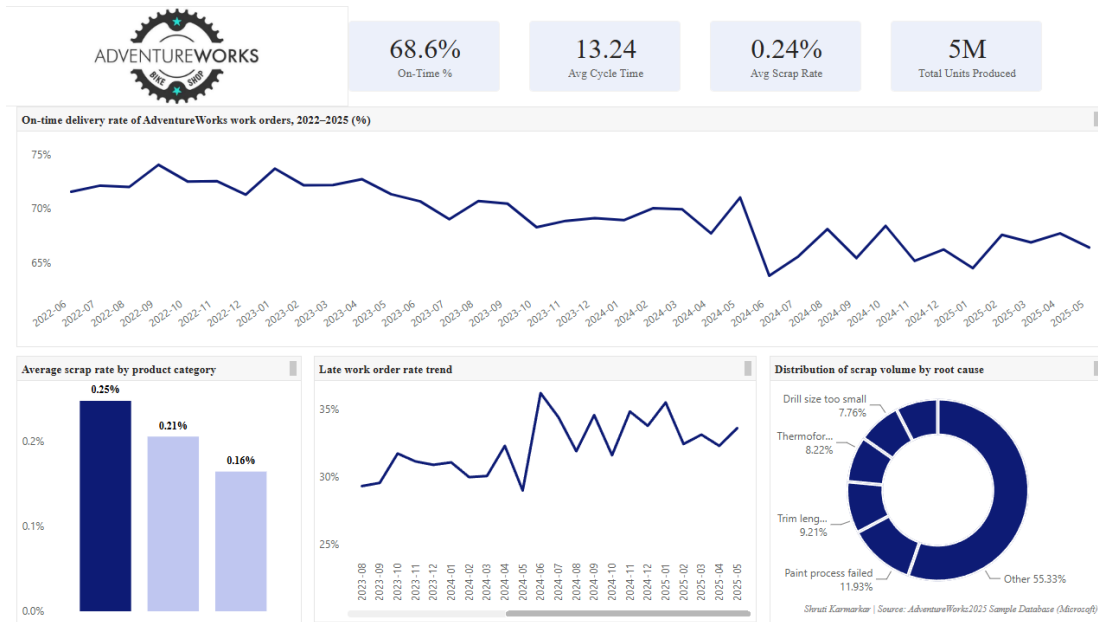
- Track on-time delivery performance and production cycle time across product categories.
- Monitor scrap rate trends and identify the root causes of driving quality loss.
- Evaluate supplier lead time, purchase order fulfillment, and vendor spend concentration.
- Identify price variance across vendors supplying the same parts, surfacing concrete cost-optimization opportunities.

Overall, this platform tells two connected stories: how efficiently AdventureWorks manufactures its products, and how effectively it manages the suppliers who support that production.

Key Findings

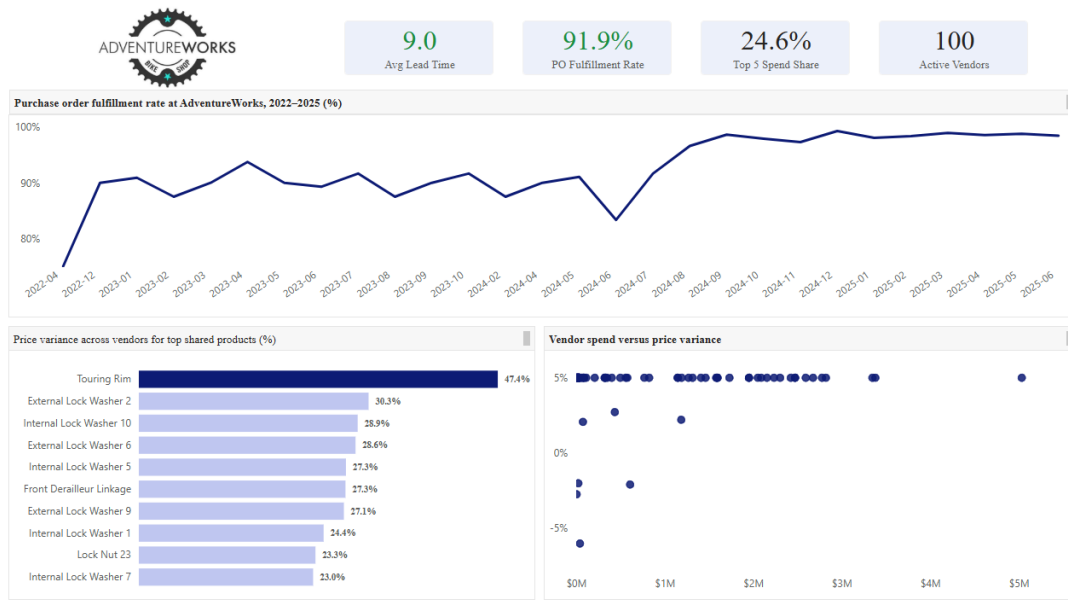
- Production volume is dominated by component-level fabrication (bearings, crankarms, races) rather than finished bike assembly these parts have no product subcategory in the source data because they are internal manufacturing inputs, not catalog items.
- Scrap is not concentrated in a single fixable root cause: the top five named reasons account for less than half of total scrap volume, with the remaining majority spread across a long tail of lower-frequency causes pointing to a broader process-discipline issue rather than one targeted fix.
- Purchase order lead time is highly consistent (~9 days) across many orders, with a small number of flagged outliers at 25 days.
- Price variance across vendors shows no relationship to vendors spend the single highest-spend vendor pays the same standard markup as low-volume vendors, while the few vendors offering genuine discounts are disproportionately small accounts. This points to an unrealized opportunity to negotiate better pricing with high-volume suppliers.

Dashboard 1: Production Operations



Production Operations — on-time delivery, cycle time, scrap rate, and root-cause analysis

Dashboard 2: Supplier & Procurement Performance



Supplier & Procurement Performance — lead time, fulfillment rate, spend concentration, and price variance

Project Repository

GitHub: <https://github.com/Shrutikarmarkar/manufacturing-ops-bi-platform>

Repository includes the full dbt project (staging, dimension, and fact models with data quality tests), the Power BI report file, and exported dashboard images.